

VMC Automatic Synthesis: Genetic Algorithm for Tendon-Driven Robotic Manipulation

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1. Introduction

- Background and motivation
 - Robot manipulation challenges
 - Virtual model control
 - motivation for automatic synthesis
- Problem statement
 - Project aims
 - Research questions
 - Scope and limitations

2. Literature Review

- Virtual model control principles and applications.
- Virtual model control design and optimization methods.
- robotic manipulation and grasping strategies.
- Genetic algorithms and other optimization techniques in robotics.
- simulation-to-real transfer and digital twin.
- gap analysis and motivation for our approach.

3. System and Infrastructure Overview

- Hardware setup and control (Sciurus17)
- VMC architecture and control flow.
- Simulation environment.
 - MuJoCo physics engine.

- model construction and simulation tooling (‘ModelBuilder’).
- contact Modeling (friction, solver, soft materials).
- tendon modeling, Jacobian mapping, and controller callbacks.
- validation and visualization tools.
- Object tracking (Aruco).

4. Automatic Synthesis Methodology

- Problem Formulation
- Parameterization of tendon configurations and controllers. (Constraining the search space)
- Objective function design and scenario randomization.
- Optimization method (Genetic Algorithm).
 - GA setup (population, selection, elitism, crossover, mutation).
 - Encoding of tendon configurations
 - Improved operators (bounded continuous mutation/crossover, integer creep).
 - Parallel evaluation and runtime considerations.
- runtime considerations and parallel evaluation.

5. Real Robot Experiments

- Initial conditions and scenario
- Evaluation metrics and data collection (digital twin, video)
- adjustments for sim-to-real transfer (contact calibration, controller scaling, jitter, ramp).
- signal processing for controller feedback. (smoothing, delay, noise handling).

6. Results

- Simulation Results
 - baseline performance (hand-designed configurations).
 - optimization progress (fitness curves, diversity).
 - best configurations and their performance.
- Real Robot controller and object tracking Results
- Simulation to real transfer results

- performance of best configurations on real robot.
- comparison to simulation results.
- qualitative observations (stability, adaptability) and unstable phenomena.
- visualization of successful grasps and failure modes.

7. Discussion

- Analysis of search space and optimization landscape. (fitness function)
- GA algorithm performance and convergency analysis.
 - effectiveness of GA operators and hyperparameter choices.
 - genome encoding and its impact on search efficiency.
 - potential improvements (adaptive mutation, multi-objective optimization).
- Analysis of discrepancies between simulation and real-world performance and how to address them.
 - Importance of accurate contact modeling and calibration.
 - joint friction and damping effects
 - controller power scaling
 - effect of multi-tendon and high damping configurations and instability.
 - sensitivity to initial conditions and robustness of controllers.
- Generalization and robustness of synthesized controllers.

10. Conclusion

- Summary of technical contributions.
- Practical impact and lessons learned. (in discussion section)

11. Future Work

- Multi-objective optimization and Pareto analysis.
- Incorporation other optimization techniques (Bayesian optimization, reinforcement learning).
- wider search space (moving sites, rotational tendons, etc).
- Better contact modeling or use of soft materials in simulation.
- making a more robust controller software stack (closed loop control, better signal processing, etc) more stable with higher damping.
- Extending to richer grasp tasks.